

Building Heaven GUI Installers for v1.0

This guide explains how to build production installers for Windows and Linux.

Prerequisites

All Platforms

- Node.js 18+ (`node --version`)
- npm 9+ (`npm --version`)
- Rust 1.70+ (`rustc --version`)

Windows

- Visual Studio Build Tools 2019 or later
- Install from: <https://visualstudio.microsoft.com/downloads/>
- Select “Desktop development with C++”
- WiX Toolset 3.11+ (for MSI installer)
- Install from: <https://wixtoolset.org/releases/>
- Add to PATH

Linux (Debian/Ubuntu)

```
sudo apt update
sudo apt install -y \
    libwebkit2gtk-4.0-dev \
    build-essential \
    curl \
    wget \
    file \
    libssl-dev \
    libgtk-3-dev \
    libayatana-appindicator3-dev \
    librsvg2-dev \
    libsoup2.4-dev \
    libjavascriptcoregtk-4.0-dev
```

Building Installers

1. Install Dependencies

```
cd heaven-gui
npm install
```

2. Build Production Bundle

```
# Build the React frontend
npm run build

# Build Tauri app with installers
npm run tauri:build
```

This will create installers in `src-tauri/target/release/bundle/` :

Windows Installers

- **MSI:** `heaven_1.0.0_x64_en-US.msi` (Windows Installer Package)
- **NSIS:** `heaven_1.0.0_x64-setup.exe` (Nullsoft Installer)

Linux Installers

- **DEB:** `heaven_1.0.0_amd64.deb` (Debian/Ubuntu package)
- **AppImage:** `heaven_1.0.0_amd64.AppImage` (Universal Linux app)

Installation Testing

Windows

MSI Installer:

```
# Install
msiexec /i heaven_1.0.0_x64_en-US.msi

# Launch from Start Menu or:
& "C:\Program Files\Heaven\Heaven.exe"

# Uninstall
msiexec /x heaven_1.0.0_x64_en-US.msi
```

NSIS Installer:

- Double-click `heaven_1.0.0_x64-setup.exe`
- Follow installation wizard
- Launch from Start Menu

Linux

DEB Package (Debian/Ubuntu):

```
# Install
sudo dpkg -i heaven_1.0.0_amd64.deb
sudo apt-get install -f # Fix dependencies if needed

# Launch
heaven-gui

# Uninstall
sudo apt remove heaven-gui
```

AppImage (Universal):

```
# Make executable
chmod +x heaven_1.0.0_amd64.AppImage

# Run directly (no installation needed)
./heaven_1.0.0_amd64.AppImage

# Optional: Integrate with system
./heaven_1.0.0_amd64.AppImage --appimage-integrate
```

Configuration

Heaven GUI reads Solo-Git configuration from:

- **Windows:** %USERPROFILE%\sologit\config.yaml
- **Linux:** ~/.sologit/config.yaml

Troubleshooting

Build Fails on Windows

Error: “WebView2 not found”

- Install WebView2 Runtime: <https://developer.microsoft.com/microsoft-edge/webview2/>

Error: “WiX not in PATH”

```
$env:PATH += ";C:\Program Files (x86)\WiX Toolset v3.11\bin"
```

Build Fails on Linux

Error: “webkit2gtk-4.0 not found”

```
sudo apt install libwebkit2gtk-4.0-dev
```

Error: “Permission denied”

```
chmod +x src-tauri/target/release/bundle/appimage/heaven_1.0.0_amd64.AppImage
```

Code Signing (Optional)

Windows

1. Obtain code signing certificate (.pfx)
2. Set environment variables:

```
powershell
$env:TAURI_SIGNING_PRIVATE_KEY = "path/to/cert.pfx"
$env:TAURI_SIGNING_PASSWORD = "your_password"
```

3. Build: Tauri will auto-sign

Linux

- DEB packages can be signed with `dpkg-sig`
- AppImages can be signed with `gpg`

Distribution

Windows

- **Recommended:** Distribute both MSI and NSIS
- MSI for enterprise/silent installs
- NSIS for regular users (smaller, faster)

Linux

- **Recommended:** Distribute both DEB and AppImage
- DEB for Debian/Ubuntu users
- AppImage for universal compatibility

Build Artifacts

After successful build, artifacts are in:

```
src-tauri/target/release/bundle/  
├── msi/  
│   └── heaven_1.0.0_x64_en-US.msi  
├── nsis/  
│   └── heaven_1.0.0_x64-setup.exe  
├── deb/  
│   └── heaven_1.0.0_amd64.deb  
└── appimage/  
    └── heaven_1.0.0_amd64.AppImage
```

Release Checklist

- [] Version updated in all files (Cargo.toml, package.json, tauri.conf.json)
- [] Icons present in `src-tauri/icons/`
- [] Build succeeds on Windows
- [] Build succeeds on Linux
- [] Windows MSI installs and launches
- [] Windows NSIS installs and launches
- [] Linux DEB installs and launches
- [] Linux AppImage runs
- [] Solo-Git CLI integration works
- [] Configuration file is read correctly
- [] AI routing system functions
- [] All UI panels load

Support

For build issues, see:

- Tauri Documentation: <https://tauri.app/v1/guides/>
- GitHub Issues: <https://github.com/ssdajoker/Solo-Git/issues>